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R-Notebook: Facial expression analysis-based blink detection

This Help Center article provides information about the facial expression analysis-based blink detection R-Notebook; it is only available for Affectiva AFFDEX.

'Blinking', the rapid closing of the eyelids, is a semi-autonomous bodily function. It serves, in the first place, the moisturizing of the eyes by spreading the tears and lubricants the eyes secrete across the surface of the eyeball. However, blinking has also been associated to psychological processes, including attention processes.

The facial expression analysis (FEA)-based blink detection R-Notebook detects and reports eye blinks based on the 'eye closure' metric provided by Affectiva AFFDEX.

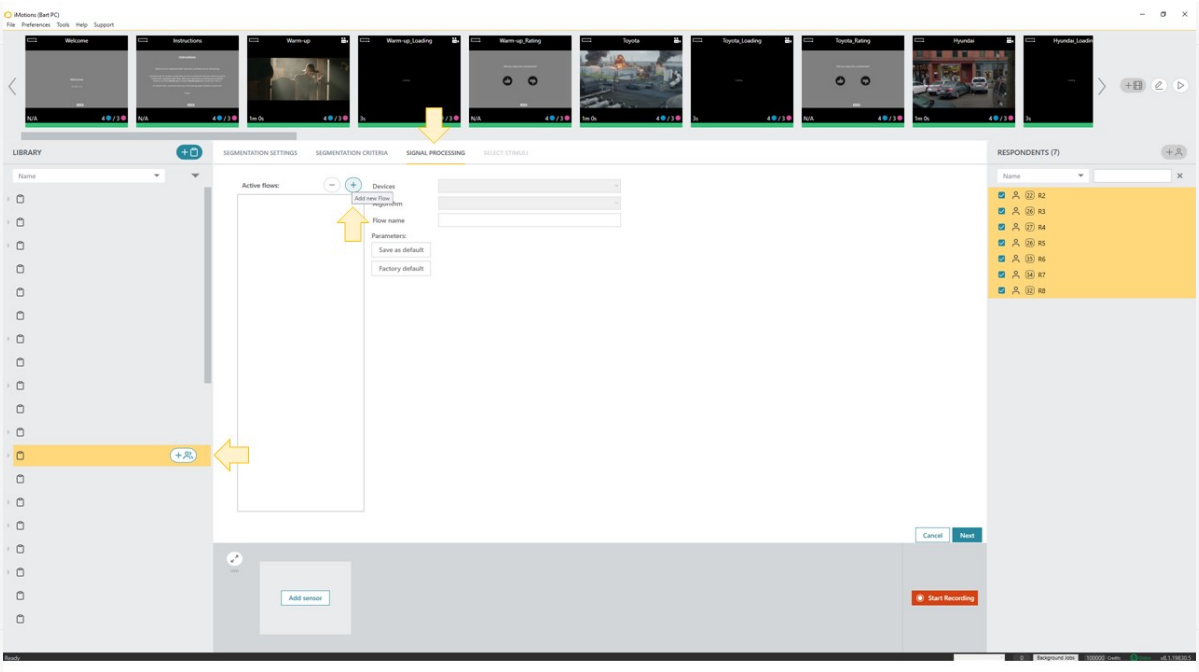
In this help center article:

- How to add the FEA-based blink detection R-Notebook to your analysis
- Details about the algorithm
- Parameters
- What the R-Notebook delivers: replay, export files, report

How to add the FEA-based blink detection R-Notebook to your analysis

The FEA-based blink detection can be added to an analysis during the initiation of a new analysis, or later, by returning to the signal processing page from within an existing analysis.

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Read more: What are R-Notebooks?

Video walkthrough

This video provides a brief walkthrough and explanation of the notebooks for both eye tracking and facial expression analysis-based blink detection:

Details about the algorithm

If you have collected data using Affectiva's AFFDEX SDK 5.1, released February 2022:

For studies collected with Affectiva's AFFDEX SDK 5.1, blink statistics are *not* computed by iMotions. Therefore, the parameters described below won't be applicable to data collected from Affectiva's AFFDEX SDK 5.1. AFFDEX SDK 5.1 provides "Blink" and "BlinkRate" directly.

Instead, the blink signal provided by Affdex is used and made available in the blink metrics export file.

To review the blink data, please refer to the blink signals from the Affdex algorithm.

Refer to *this article* to know more about changes in AFFDEX SDK 5.1.

If you have collected data using Affectiva's AFFDEX SDK 4.0:

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The algorithm is based on the ‘eye closure’ metric provided by Affectiva AFFDEX. This metric expresses the likelihood of the eyes being closed on a scale from 0 to 100; this could be roughly understood as the *probability* that the eyes are closed.

The algorithm then simply tags blinks if the eye closure values exceed a given probability threshold (see parameter: probability threshold). Specifically, a blink is tagged from the moment that the eye closure values first exceed the threshold until the eye closure values drop below the threshold again.

These tags are then used to count the number of blinks and to compute the blink rate (i.e., number of blinks per minute) and interblink intervals.

Parameters

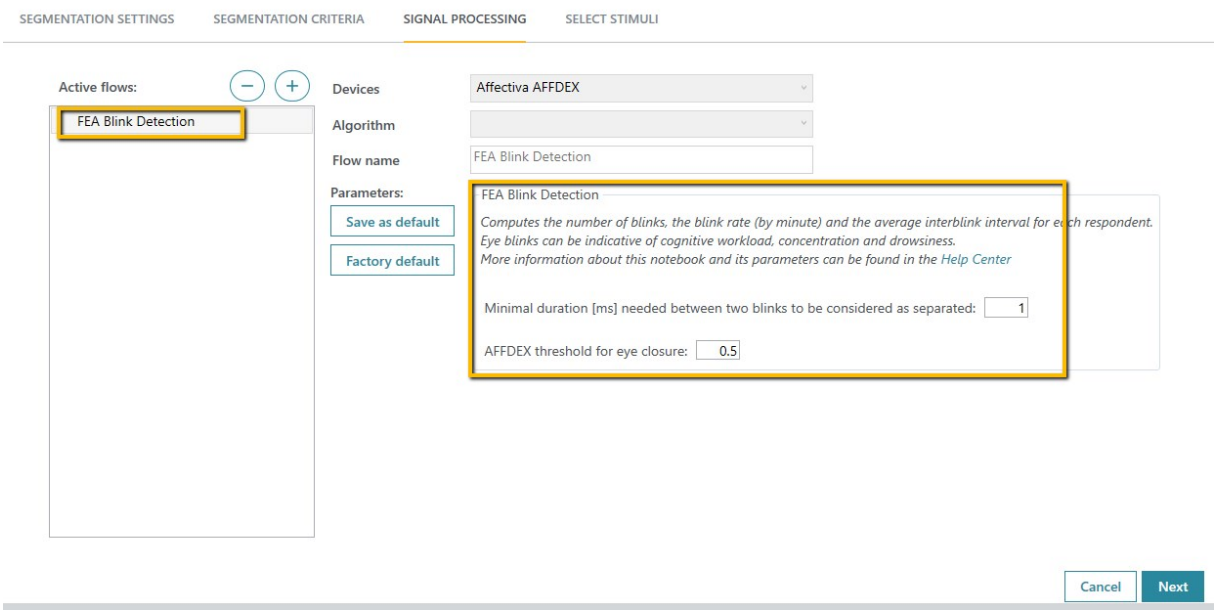
The sensitivity of the blink detection R-Notebook can be altered by changing the following two parameters:

- **Minimum duration [ms] needed between two blinks to be considered as separated**: The minimum duration between two blinks in milliseconds (i.e., from the last sample-above-threshold of the first blink to the first sample-above-threshold of the next blink.
- AFFEDEX threshold for eye closure: The threshold used for detecting a blink in the ‘eye closure’ data

The defaults are set to:

- **Minimum duration [ms] needed between two blinks to be considered as separated**: 1 millisecond
- **AFFEDEX threshold for eye closure**: 0.5

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This probability threshold of 0.5 (which means 0.5%, not 50%) seems low, but was chosen because it rendered the most optimal results during our in-house testing. It combines a maximal sensitivity of the blink detection with a negligible amount of false positives.

Also, note that the 'minimum duration between two blinks' is set to 1 millisecond. This means that a single data sample below the threshold is sufficient to separate two consecutive eye blinks. It often happens that respondents make two blinks in fast succession; in that case, the 'eye closure' values sometimes do not even drop below the threshold in between the two blinks - or only very briefly. Setting the 'minimum duration between two blinks' to 1 millisecond maximally separates these double-blinks (i.e., it counts double-blinks as two blinks instead of one). However, by increasing this value, you can choose not to separate those double-blinks (i.e., always count double-blinks as one).

For that reason, the algorithm will not always be able to count these blinks as separate blinks; the algorithm counts them as a single blink.

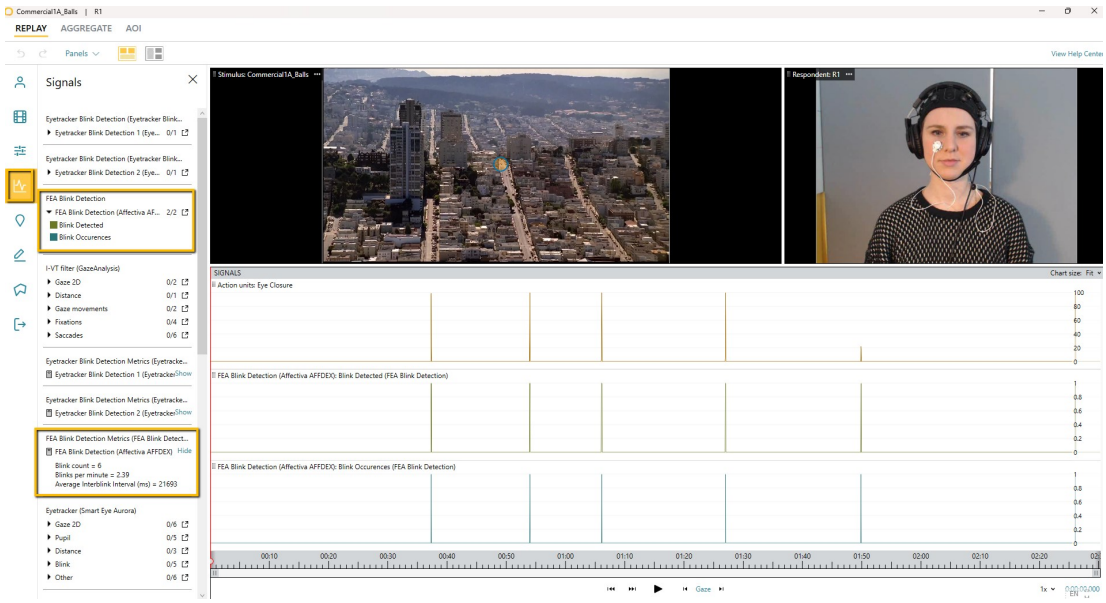
What the R-Notebook delivers: replay, export files, report

The FEA-based blink detection R-Notebook performs detection of eye blinks and further computes (1) the total number of blinks during a stimulus, scene and/or annotation, and (2) the corresponding blink rate during the same time-windows (i.e., the number of blinks per minute), (3) the average of interblink intervals.

- Replay: The R-Notebook adds two sections to the signals page in the Replay: ‘Blink Detection’ and ‘Blink Detection Metrics’. The section ‘Blink

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Detection' (AFFDEX SDK 4.0 only) contains a binary signal that can be used for visualizing the blinks on the timeline (Blink Detected) and another binary signal that can be used for visualizing the maximum value of the "Eye Closure" metrics of each blink on the timeline (Blink Occurrences). The section 'Blink Detection metrics' further provides two quick statistics – the total number of blinks (Blink count), blink rate (Blinks per minute), and average of interblink intervals (Average Interblink Intervals (ms)) for the respondent and stimulus currently displayed. Note that, here, this is only the blink count, blink rate and interblink interval for the *entire* stimulus duration.

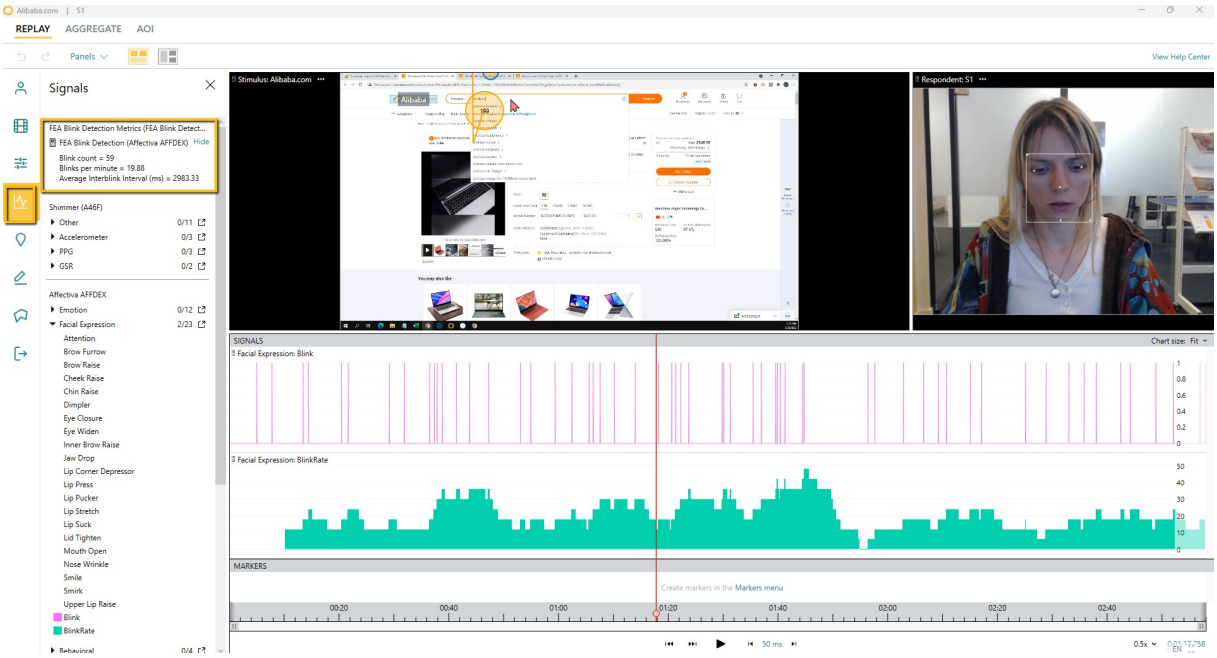


(Data using Affectiva's AFFDEX SDK 4.0)

If you have collected data using Affectiva's AFFDEX SDK 5.1:

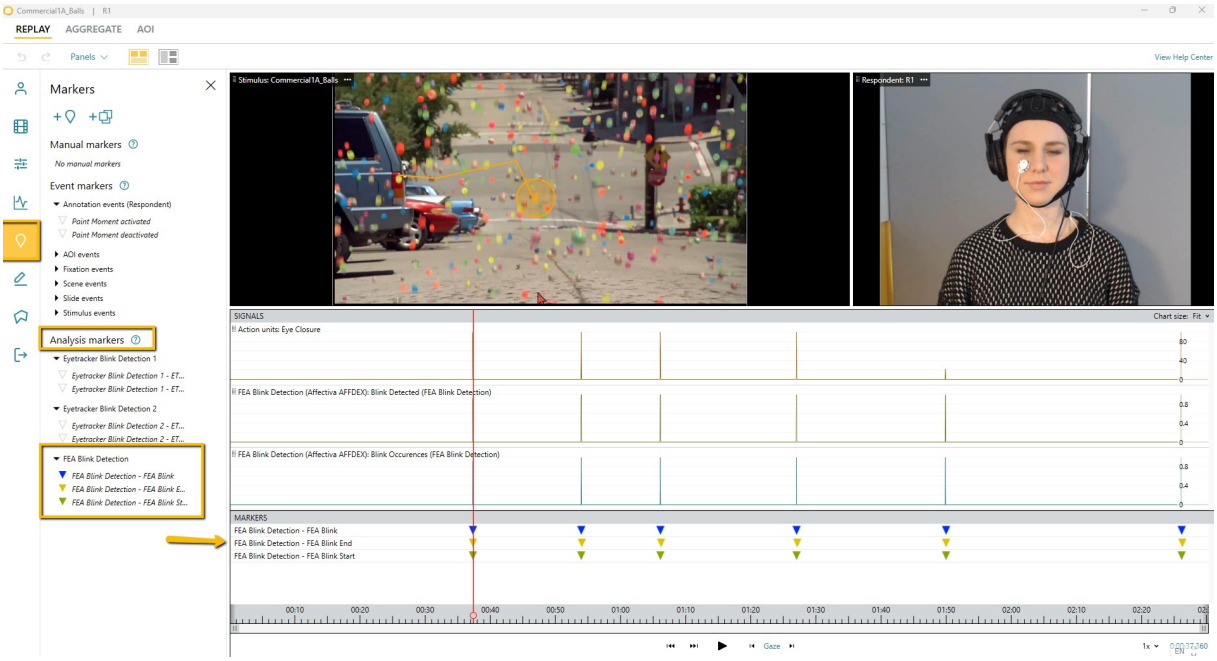
You will only have the "Blink Detection Metrics" displayed. You can expand the "Blink" and "BlinkRate" metrics provided by AFFDEX SDK 5.1 to visualize where the blinks are detected.

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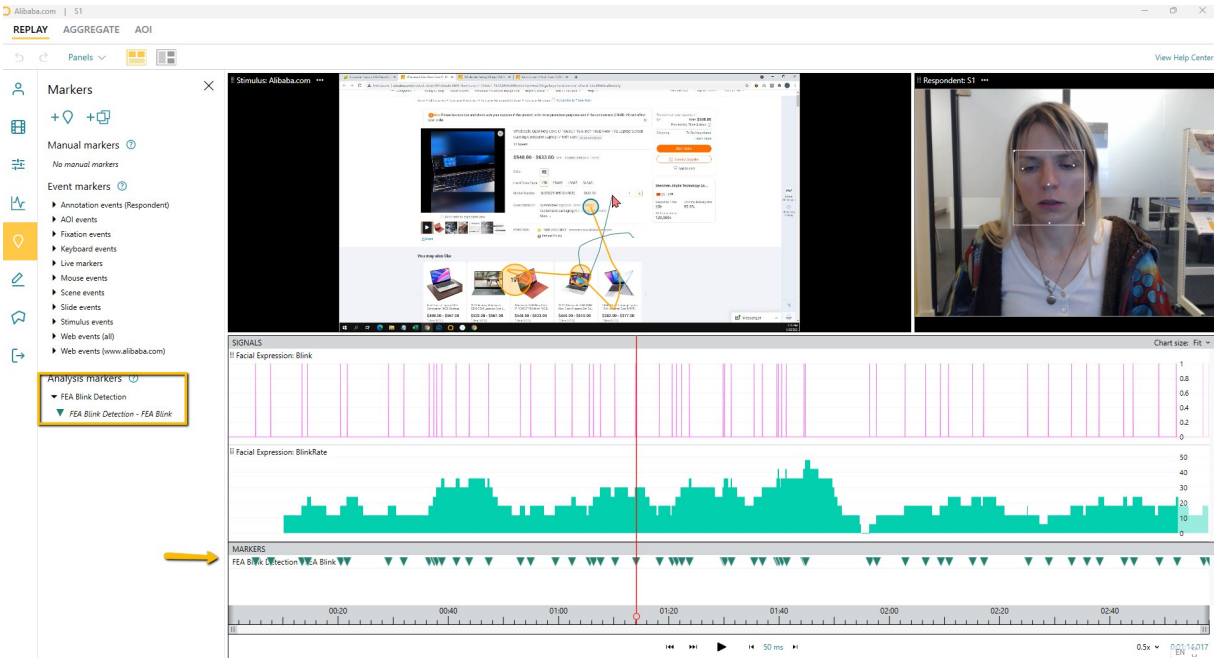
(Data using Affectiva's AFFDEX SDK 5.1)

Now in the replay window, you will also be able to obtain Analysis markers generated by the R Notebook. Please see the screenshot below as an illustration of where to find them. The Analysis markers for FEA Blink Detection R Notebook are composed of "FEA Blink Start" (AFFDEX SDK 4.0 only), "FEA Blink" and "FEA Blink End" (AFFDEX SDK 4.0 only). Each markers section is named after the name of the "Active flows" user-defined when setting up Analysis. Users can also create Automated annotations based on these markers. Please see this article for more guides on how to create Automated annotations.



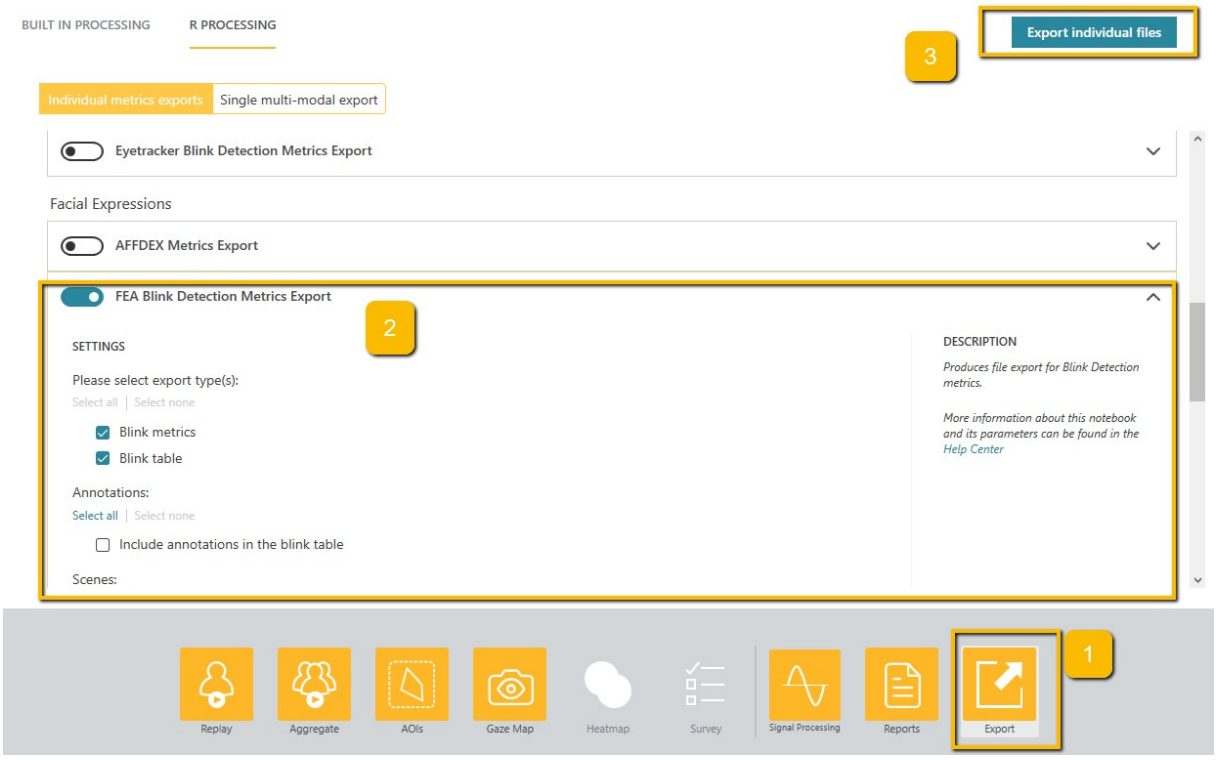
(Data using Affectiva's AFFDEX SDK 4.0)

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(Data using Affectiva's AFFDEX SDK 5.1)

Export blink metrics: The R-Notebook also produces two types of export files: FEABlinkMetrics and FEABlinkTable.



File FEABlinkMetrics:

File FEABlinkMetrics, for *every* respondent included in the analysis and for *every* stimulus, scene, or annotation present in the study (each row), lists the corresponding number of blinks (Blink count), Blinks Rate and Interblink Interval.

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#METADATA												
#Channel	Info_Resp	Info_Resp	Info_Resp	Info_Resp	Info_Inter	Info_Inter	Info_Inter	Info_Inter	Info_Inter	FEA_BlinkMetrics_Blin	FEA_BlinkMetrics_Blin	FEA_BlinkMetrics_InterblinkInterval
#Group	Responde	Responde	Responde	Responde	Interval Ir	Interval Ir	Interval Ir	Interval Ir	Interval Ir	Blink Metrics	Blink Metrics	Blink Metrics
#Unit							Millisecor	Milliseconds				Milliseconds
#Short unit							ms	ms				ms
#Descripti	Name of t	Gender of	Age of the	Group of t	Type of th	Name of t	Start time	Duration (	Name of t	Count of all blinks det	Blink rate during this i	Average time interval between blinks.
#Device										Affectiva AFFDEX	Affectiva AFFDEX	Affectiva AFFDEX
#AlgorithmType										FEA Blink Detection	FEA Blink Detection	FEA Blink Detection
#Flow Name										FEA Blink Detection	FEA Blink Detection	FEA Blink Detection
#DATA												
Study Nan	Responde	Gender	Age	Group	Type	Label	Start (ms)	Duration (	Parent S	Blink Count	Blink Rate	Interblink Interval
SonyBravi	R1	FEMALE	27	Default	Stimulus	Commerci	89361	150341		6	2.39	21693
SonyBravi	R1	FEMALE	27	Default	Annotatio	Balls	119604	31896	Commerci	2	3.76	16593
SonyBravi	R1	FEMALE	27	Default	Stimulus	Commerci	248392	71144		8	6.75	9634.43
SonyBravi	R2	FEMALE	23	Default	Stimulus	Commerci	82950	71411		27	22.69	2576.62
SonyBravi	R3	MALE	30	Default	Stimulus	Commerci	260117	71405		4	3.36	19498.33
SonyBravi	R4	FEMALE	26	Default	Stimulus	Commerci	86492	150418		109	43.48	1232.01
SonyBravi	R4	FEMALE	26	Default	Stimulus	Commerci	249190	71496		51	42.8	1256.68

File FEABlinkTable:

File FEABlinkTable, for every detected blink (each row), lists the corresponding Blink number, counting from start of the recording (Blink Index), Blink number, counting from start of the stimulus (Blink Index by Stimulus), Start time of the blink (relative to the recording start) in Milliseconds (Blink Start) (AFFDEX SDK 4 only), Time of the blink when the maximum value of the "FEA" signal was reached (relative to recording start) (Blink Time), End time of the blink (relative to recording start) in Milliseconds (Blink End)(AFFDEX SDK 4 only), Time interval between two subsequent blinks in Milliseconds (Interblink Interval), and the corresponding hardware used, Active flows the blink belongs to. If the "Annotations" or "Scenes" were clicked when exporting from iMotions, there will be added columns of Scenes and Annotations indicating whether the blink falls into them or not.

#METADATA																								
#Channel	Info_Resp	Info_Resp	Info_Resp	Info_Resp	Info_Inter	Info_Inter	Info_Inter	FEA_Blinks_BlinkIndex			FEA_Blinks_BlinkIndexbyStimulus			FEA_Blinks_B	FEA_Blinks_Bl	FEA_Blinks_I	FEA_Blinks_Interblir	FEA_Blink	FEA_Blinks_Flc	Annotations_Balls_Active				
#Unit					Millisecor	Millisecor	Millisecor	Index			Index			Milliseconds	Milliseconds	Milliseconds	Milliseconds							
#Short unit					ms	ms								ms	ms	ms	ms							
#Descripti	Name of t	Gender of	Age of the	Group of t	Name of t	Start time	Duration (	Blink number, counting from start of the recording.			Blink number, counting from start of the stimulus.			Start time of	Time of the bl	End time of t	Time interval betwe	Name of t	User-defined r	Indicates when Annotation				
#DATA																								
Study Nan	Responde	Responde	Responde	Responde	Stimulus I	Stimulus S	Stimulus	Blink Index			Blink Index by Stimulus			Blink Start	Blink Time	Blink End	Interblink Interval	Device	Flow Name	Balls active				
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							1	126687	126719	126751	38400	Affectiva	FEA Blink Dete	1			
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							2	143344	143376	143424	16593	Affectiva	FEA Blink Dete	1			
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							3	155456	155520	155552	12032	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							4	176384	176416	176448	20832	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							5	199312	199312	199344	22864	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	89360.74	150341.4							6	235488	235519	235584	36144	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							1	249047	249094	249094	6771	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							2	265318	265446	265526	16224	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							3	268119	268150	268150	2593	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							4	276550	276550	276582	8400	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							5	312950	313014	313078	36368	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							6	313958	313990	314086	880	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							7	316950	316982	317078	2864	Affectiva	FEA Blink Detection				
SonyBravi	R1	FEMALE	27	Default	Commerci	248391.7	71144.49							8	317190	317254	317318	112	Affectiva	FEA Blink Detection				
SonyBravi	R2	FEMALE	23	Default	Commerci	82949.56	71410.6							1	84340	84372	84420	1929	Affectiva	FEA Blink Detection				

(Data using Affectiva's AFFDEX SDK 4.0)

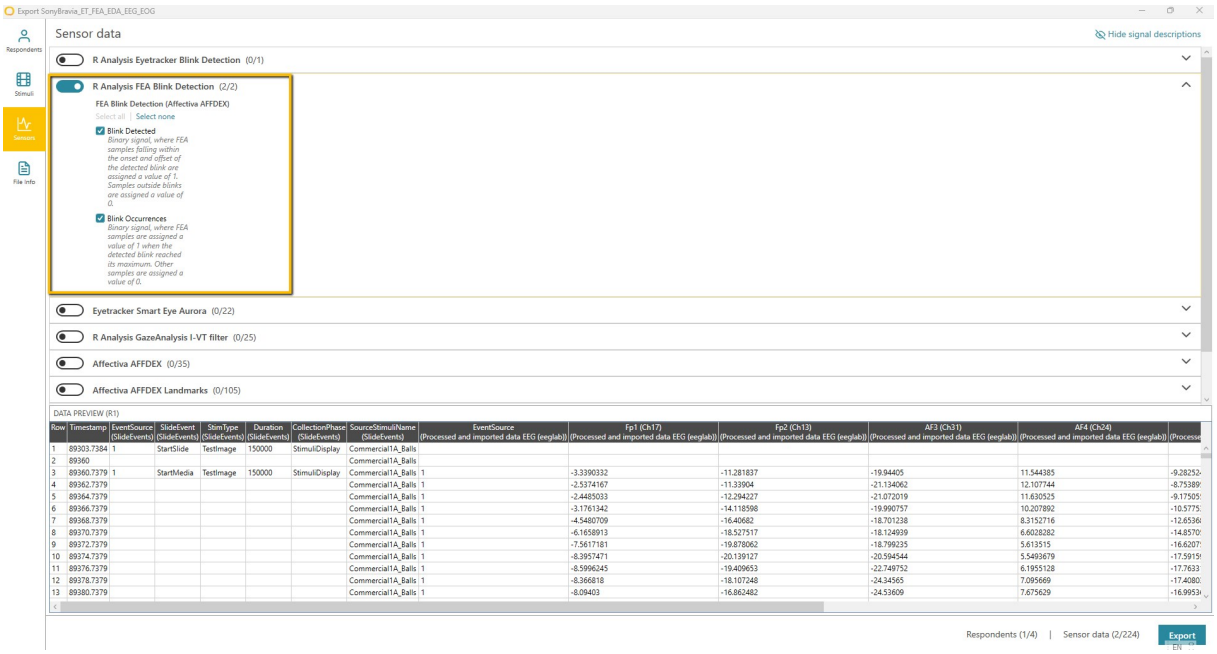
#METADATA																								
#Channel	Info_Resp	Info_Resp	Info_Resp	Info_Resp	Info_Inter	Info_Inter	Info_Inter	FEA_Blinks_BlinkI			FEA_Blinks_BlinkIndexbyS			FEA_Blinks_B	FEA_Blinks_Interblir			FEA_Blink	FEA_Blinks_Flc	Scenes_Amaz	Annotatio	Annotat		
#Unit					Millisecor	Millisecor	Index				Index			Milliseconds	Milliseconds									
#Short unit					ms	ms								ms	ms									
#Descripti	Name of t	Gender of	Age of the	Group of t	Name of t	Start time	Duration (	Blink number, cou			Blink number, counting fro			Time of the b	Time interval betwe			Name of t	User-defined r	Indicates whe	Indicates	Indicate		
#DATA																								
Study Nan	Responde	Responde	Responde	Responde	Stimulus I	Stimulus S	Stimulus	Blink Index			Blink Index by Stimulus			Blink Time	Interblink Interval			Device	Flow Name	Amazon gaze	0 Task exe	1 First t		
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	32			1			49536	4611			Affectiva	AFFDEX 5.1		1			
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	33			2			52236	2700			Affectiva	AFFDEX 5.1		1			
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	34			3			56436	4200			Affectiva	AFFDEX 5.1		1			
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	35			4			59436	3000			Affectiva	AFFDEX 5.1	1	1			
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	36			5			63670	4234			Affectiva	AFFDEX 5.1	1	1			
WebUX(e-S1		FEMALE	32	Default	Amazon.c	46982.5	49156.13	37			6			65670	2000			Affectiva	AFFDEX 5.1	1	1			

(Data using Affectiva's AFFDEX SDK 5.1)

- Export sensor data: It is further possible to export the binary blink

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detection data (Blink Detection) as part of the sensor data export file. This is done by enabling the following in the "Sensor Data" Export (AFFDEX 4.0 only):



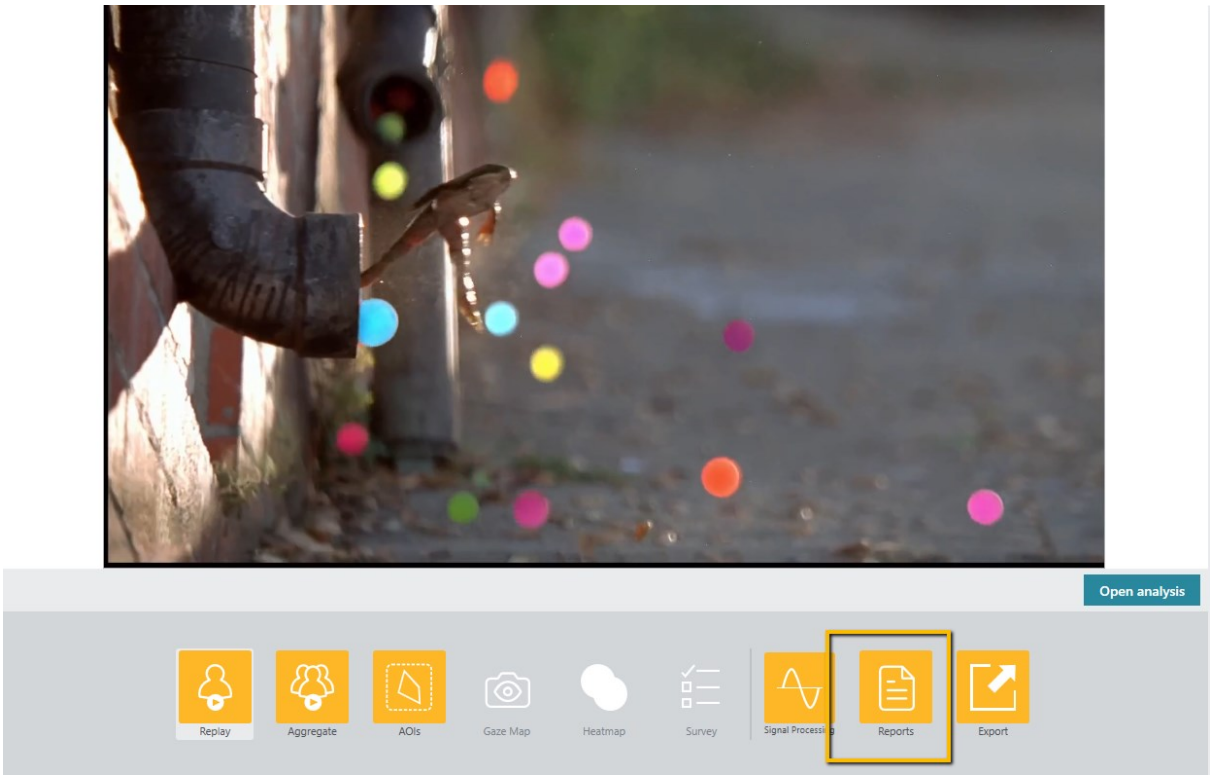
(Data using Affectiva's AFFDEX SDK 4.0)

Read more: Export: Sensor data

If you wish to review the R-code:

- Report: By clicking into "Report" button you will be able to view a report with the R-code embedded and a step-by-step description of the algorithm.

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FEA Blink Detection

Code

Show
Show
Show
Show

Study: `SonyBravia_ET_FEA_EDA_EEG_EOG`
Respondent: `R1`
Sensor: `Affectiva AFFDEX 4`

Methods

This R notebook has performed the following steps:

- Blink tagging when the FEA eye closure is superior to a threshold of 0.5.
- Merger of two separate blinks when the time duration between them is inferior to 1 ms (deal with fluctuation in eye closure probability). Note that, by default, no blinks are merged.
- Determination of the interblink interval by calculating the time interval between two subsequent blinks.
- Computation of the number of blink, the blink rate by minute and the average interblink interval.

Show

FEA blink results

Show

Commercial1A_Balls | Commercial2_Paint

Blinks metrics

Number of blinks: 6

Blinks per minute: 2.39

Average Interblink Interval: 21693.00 ms

Blinks table

Note: Blinks are given by stimulus and the start/end time of blinks (relative to stimulus start) falling outside the given stimulus are set to NA.

Blink Start	Blink	Blink End	Interblink Interval (ms)
37326.26	37358.26	37390.26	NA
53983.26	54015.26	54063.26	16593
66095.26	66159.26	66191.26	12032
87023.26	87055.26	87087.26	20832
109951.26	109951.26	109983.26	22864
146127.26	146158.26	146223.26	36144

Show

Show

Computation started at 2023-11-02 12:04:36 CET / Notebook execution time: 3.898511 secs

The FEA-based blink detection R-Notebook was released as part of iMotions version 8.1.6

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